

# **COMP1008 - Fundamentals of AI**

## **Knowledge Representation & Reasoning**

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# Knowledge Based Systems

- ▶ Humans use not only **methods**, but also domain **knowledge**, in problem solving
  - Facts, concepts, procedures, models, examples, etc. about the world
  - Dominate AI method till 90's
  - Hard-coded manually by experts
  - Interpretable
  - Question: can all tasks be formalised?
- How to deal with **vague and uncertain** knowledge?
  - Probability: Bayes Rule
  - Fuzzy logic

| AI                      | COMP1008      | CS Course |
|-------------------------|---------------|-----------|
| Deduction<br>(Top-Down) | KRR           | COMP3008  |
| Deduction               | Fuzzy Systems | COMP4033  |

# Knowledge Based Systems

- ▶ A system built around a **knowledge base**. i.e. a collection of knowledge, **taken** from a human, and **stored** in such a way that the system can **reason** with it
  - Substitute, to certain extent, the experts
  - Make good decisions quickly
  - **Domain specific**: a narrow range of knowledge
- Expert systems
  - Knowledge base: a list of **rules**
  - E.g. **if** TV spent = 3k, **then** Sales = 3.6k
- Case based reasoning systems
  - Knowledge base: a list of **cases**
  - E.g. in **case A**, TV spent of 3k lead to Sales of 3.6k
- New problem: TV spent 4k, predicted Sales = ?

| AI               | Input                   | Approach           | Output        |
|------------------|-------------------------|--------------------|---------------|
| Machine learning | Data (input and output) | Training algorithm | Models        |
| KBS              | Facts / knowledge       | Reasoning          | New knowledge |

# Knowledge Based Systems

- ▶ A system built around a **knowledge base**. i.e. a collection of knowledge, **taken** from a human, and **stored** in such a way that the system can **reason** with it
- 1. **Acquire** knowledge
  - transfer expertise from knowledge source to a program
- 2. **Represent** knowledge
  - symbolic encoding of propositions
- 3. **Apply** the knowledge, i.e. **reasoning**
  - deduce logical consequences: to produce *new* propositions

# KBS: Acquisition / Engineering

- **Bottleneck** of building a knowledge-based system
  - Human knowledge is complex, unstructured and usually ill formulated
  - Experts often too subjective, knowledge may even conflict
  - Iterative procedure, time consuming to maintain
  - Missing knowledge?

1. Define task
2. Build domain **Vocabulary**, i.e. words, phrases, formulae
3. Develop a Model of the **Reasoning** and how it is applied
  - Flowcharts and decision trees often used
  - Paper exercise

```
Welcome to

EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LL      II    ZZZ   AAAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LLLLLL  IIII  ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

# KBS: Representation

- ▶ Natural language to represent and handle facts
  - spot is a dog
  - all dogs have tails
- ▶ Then deduce:
  - spot has a tail
- ▶ Problems with natural language
  - Often ambiguous
  - Little **uniformity** in the structure of sentences



# KBS: Representation – predicate logic

- ▶ A symbolic formal language to represent knowledge
    - Expressive to derive *new* knowledge from old, through mathematical deduction
    - Consider the following **fact**  
“spot is a dog”: **dog**(spot)
    - How to represent “all dogs have tails”?
  - Proposition: connected **predicates** (of facts on variables), refined by **quantifiers**
  - **Fact**: dog (variable) has a tail: **hastail**(x)
  - **Quantifier**: all dogs, universally:  $\forall$
  - $\forall x \text{ dog}(x) \rightarrow \text{hastail}(x)$
- |               |                           |
|---------------|---------------------------|
| $\forall$     | for all                   |
| $\exists$     | there exists              |
| $\vee$        | disjunctive (logical or)  |
| $\wedge$      | conjunctive (logical and) |
| $\neg$        | negate (logical not)      |
| $\rightarrow$ | implication (implies)     |

|                       |            |                                  |
|-----------------------|------------|----------------------------------|
| Variables & Constants | Predicates | Connectives & Quantifiers        |
| Objects               | Facts      | Reasoning with defined variables |

# KBS: Representation – past exam questions

► Represent the following by propositions (3 marks)

- “All animals eat fruit or meat”

$\text{animal}(x)$

$\text{eat}(x, \text{fruit}) \vee \text{eat}(x, \text{meat})$

$\forall x \text{ animal}(x) \rightarrow \text{eat}(x, \text{fruit}) \vee \text{eat}(x, \text{meat})$

► Try another one

- Elephants are not carnivores

|               |                           |
|---------------|---------------------------|
| $\forall$     | for all                   |
| $\exists$     | there exists              |
| $\vee$        | disjunctive (logical or)  |
| $\wedge$      | conjunctive (logical and) |
| $\neg$        | negate (logical not)      |
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|                       |            |                                  |
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# KBS: Reasoning – resolution rule

- ▶ An inference method which infer new knowledge
- ▶ Method: show the counter-example leads to a conflict in the KB (becomes empty)
- ▶ Requirement: knowledge base is in **Conjunctive Normal Form (CNF)**, i.e. propositions with clauses connected by  $\vee$

$$A \Rightarrow B \rightarrow \neg A \vee B$$

Input: a **knowledge base (KB)** and a **statement** (new knowledge we want to infer)

1. Convert the KB into **CNF**
2. **Negate** the statement, **add** it to the KB
3. **Unification**: replace variables by a concrete instance (constant)
4. **Resolution**: iteratively eliminating contradictions in the KB till
  - An empty clause is resolved (the negated statement must be false)  $\rightarrow$  original is true, or
  - No more contradictions exist  $\rightarrow$  negated statement is true

# KBS: Reasoning – resolution example

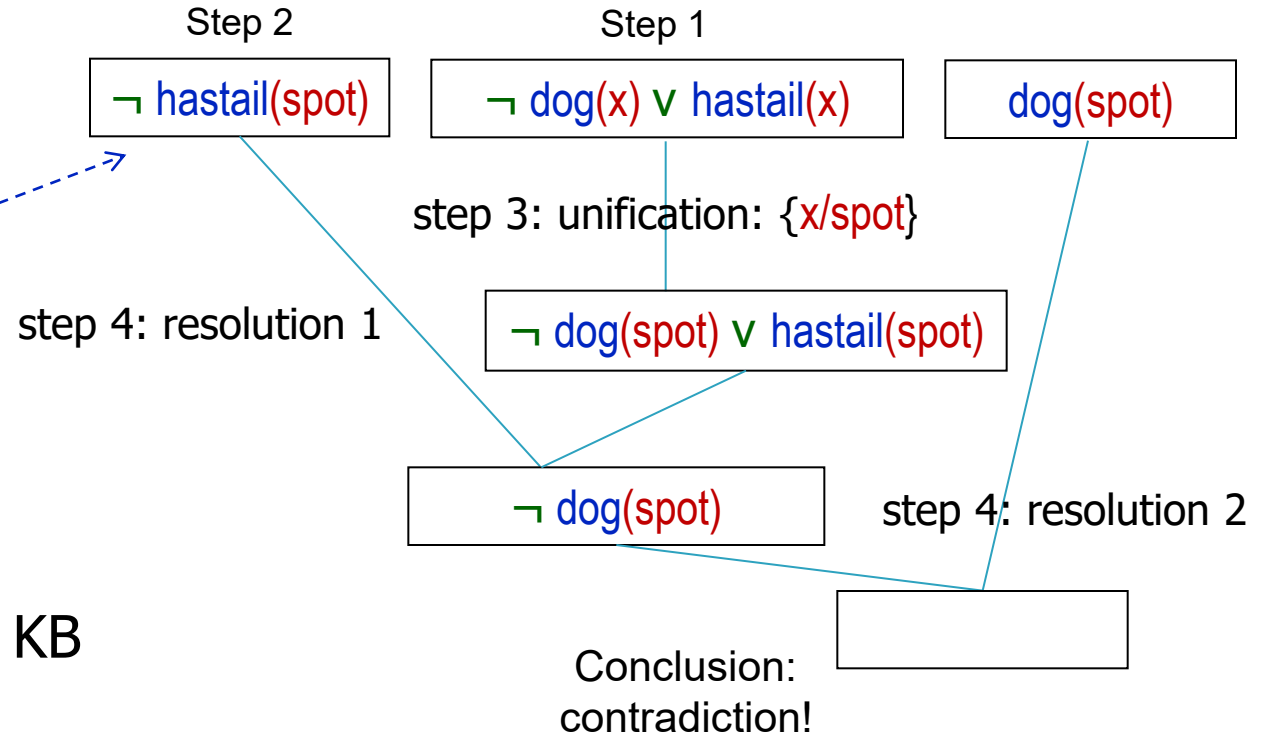
- Input: in the KB

$\forall x \text{ dog}(x) \rightarrow \text{hastail}(x)$

$\text{dog}(\text{spot})$

- Statement:  $\text{hastail}(\text{spot})$

- Step 1: Convert  $\rightarrow$  to CNF
- Step 2: add negated statement to KB
- Step 3: perform unification
- Step 4: resolution



Therefore,  $\text{hastail}(\text{spot})$  must be true, since  $\neg \text{hastail}(\text{spot})$  was proven false through resolving a contradiction

# KBS: Reasoning – past exam question

- Input: in the knowledge base

elephant(Ellie)

$\forall x \text{ elephant}(x) \rightarrow \text{animal}(x)$

$\forall x \text{ animal}(x) \rightarrow \text{eats}(x, \text{fruit}) \vee \text{eats}(x, \text{meat})$

- Statement  $\text{eats}(\text{Ellie}, \text{fruit})$

- Step 1: Convert  $\rightarrow$  to CNF
- Step 2: add negated statement to KB
- Step 3: perform unification
- Step 4: resolution

